Chapter 3: Motion Reference Sheet

Position, Velocity, and Acceleration **Position** tells where an object is. **Velocity** tells how fast and in what direction position is changing.

$$\boxed{v = \frac{\Delta x}{\Delta t} = \frac{\text{change in position}}{\text{change in time}}}$$

Velocity is the **slope of a position vs. time graph**.

- Positive velocity: motion in the positive direction
- Negative velocity: motion in the negative direction
- Zero velocity: the object is not moving

Acceleration tells how quickly velocity is changing.

$$\boxed{a = \frac{\Delta v}{\Delta t} = \frac{\text{change in velocity}}{\text{change in time}}}$$

Acceleration is the **slope of a velocity vs. time graph**.

Slope

$$\boxed{\text{slope} = \frac{\text{vertical change}}{\text{horizontal change}}}$$

Graph	Slope represents
position vs. time	velocity
velocity vs. time	acceleration

Average Velocity

$$\boxed{v_{\text{avg}} = \frac{\text{displacement}}{\text{time}}}$$

Equivalently,

$$\boxed{\text{displacement} = v_{\text{avg}}\,\Delta t}$$

Relative Velocity and Vector Addition When an object moves through moving water, air, or another moving medium, **add the velocity vectors**.

$$\boxed{\vec{v}_{\text{object/ground}} = \vec{v}_{\text{object/medium}} + \vec{v}_{\text{medium/ground}}}$$

- Vectors pointing in the **same direction** add.
- Vectors pointing in **opposite directions** subtract.
- For perpendicular vectors, draw a right triangle.

For a right triangle,

$$\boxed{a^2 + b^2 = c^2}$$

Units and Symbols

Quantity	Symbol	Unit
position / displacement	$x,\ y,\ \Delta x$	m
velocity	v	m/s
acceleration	a	m/s^2

The symbol Δ means **change in**:

$$\Delta x = x_{\text{final}} - x_{\text{initial}} \qquad \Delta v = v_{\text{final}} - v_{\text{initial}}$$

Pay attention to signs and directions. A negative velocity or acceleration does **not** automatically mean "slowing down."